



MARKET RETAIL ANALYTICS PROJECT-ML2

GROCERY RETAIL MARKET BASKET ANALYSIS

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PROBLEM STATEMENT



In the highly competitive grocery retail industry, maintaining a strong customer base and ensuring profitability require a deep understanding of customer buying behavior. As the market continues to evolve and consumer preferences change, grocery retailers must continually refine their strategies to stay ahead of the competition.

One of the most powerful ways to achieve this is by identifying and analyzing the patterns in customer purchasing behavior, particularly the combinations of products frequently bought together.

The core challenge faced by grocery retailers is how to use vast amounts of transactional data collected through Point of Sale (POS) systems to uncover actionable insights.

Understanding which items customers commonly purchase together enables stores to identify opportunities for targeted marketing, better inventory management, and tailored promotions that resonate with customers' preferences.

This analysis aligns with business goals by maximizing revenue, reducing operational costs, and boosting customer loyalty.

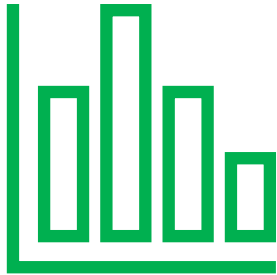
KEY OBJECTIVES



- Identifying frequently purchased item combinations.
- Extracting patterns from POS transactional data using Association Rule Mining.
- Calculating and evaluating metrics like support, confidence, and lift to ensure strong, meaningful relationships between products.
- Uncovering patterns that will help the store create targeted combo offers and discounts.
- Obtaining insights for driving revenue growth by increasing customer purchases and average basket size.

**The insights derived from Association Rule Mining can be used to guide personalized marketing strategies for improved sales and customer retention.*

DATA OVERVIEW



The transactional data from a grocery store, comprising 20641 rows and 3 columns, where each row represents a product purchased in a specific order.

The key attributes include:

- **Date:** The date when the transaction took place.
- **Order_id:** A unique identifier for each customer order.
- **Product:** The individual item purchased in the transaction.

EXPLORATORY DATA ANALYSIS



Summary Statistics

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20641 entries, 0 to 20640
Data columns (total 3 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Date         20641 non-null  datetime64[ns]
1   Order_id     20641 non-null  int64
2   Product      20641 non-null  object
dtypes: datetime64[ns](1), int64(1), object(1)
memory usage: 483.9+ KB
```

	count	unique	top	freq	mean	min	25%	50%	75%	max	std
Date	20641	NaN	NaN	NaN	2018-12-21 14:03:01.735381248	2018-01-01 00:00:00	2018-05-29 00:00:00	2019-01-30 00:00:00	2019-06-20 00:00:00	2020-02-26 00:00:00	NaN
Order_id	20641.0	NaN	NaN	NaN	575.986289	1.0	292.0	581.0	862.0	1139.0	328.557078
Product	20641	37	poultry	640	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Inferences

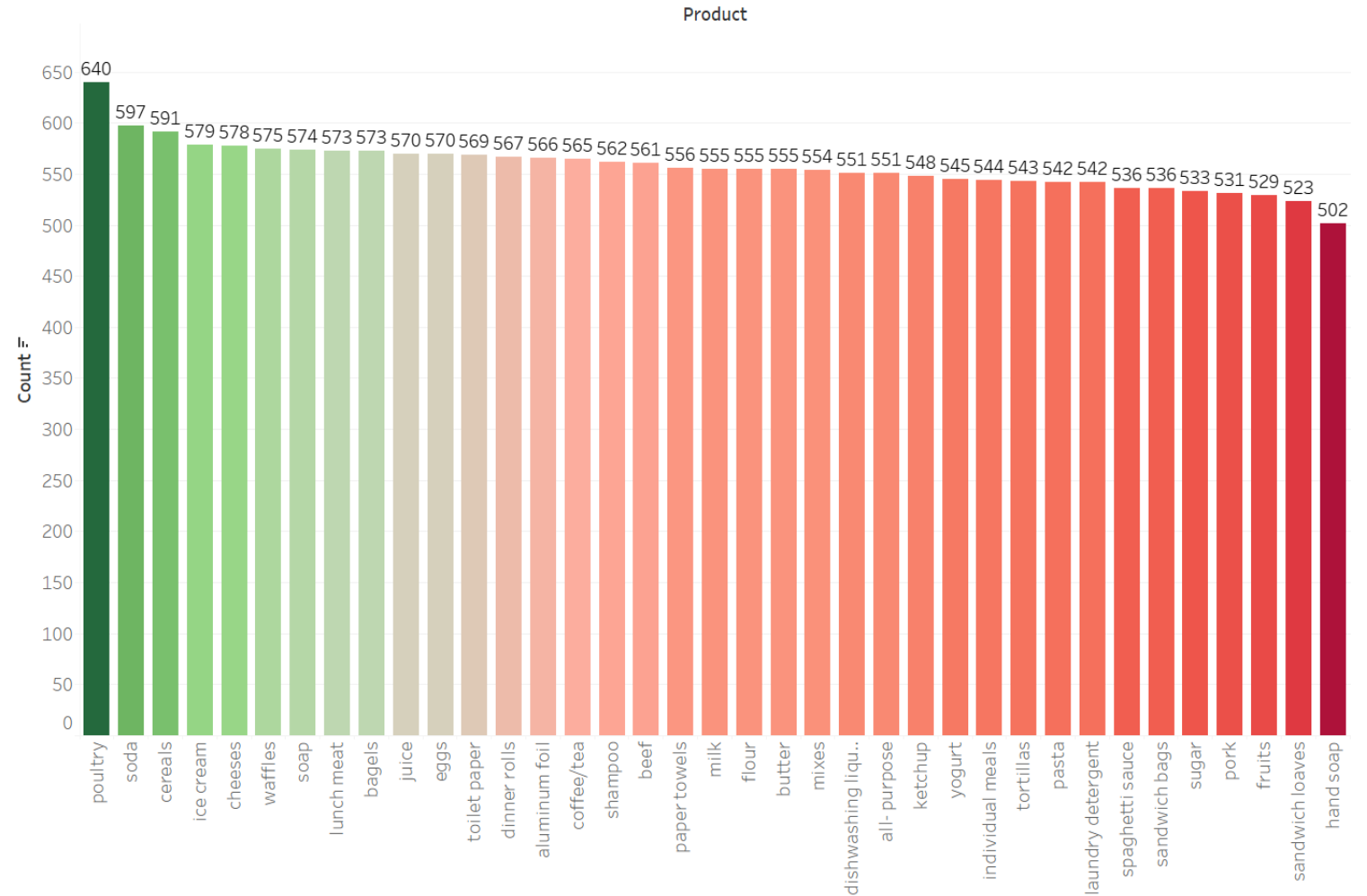
- The dataset contains a total of 20,641 transactions.
- There are no null values in the data.
- There are 1139 unique orders and 37 unique products, among which poultry has the highest count, with 640 occurrences.

1. UNIVARIATE ANALYSIS

1.1 Frequency Distribution of Products

Poultry has the highest count in the transaction order with 640, followed closely by Soda (597), Cereals (591), Ice Cream (579), and Cheeses (578).

Although Poultry ranks first, the counts of the following products are quite similar.

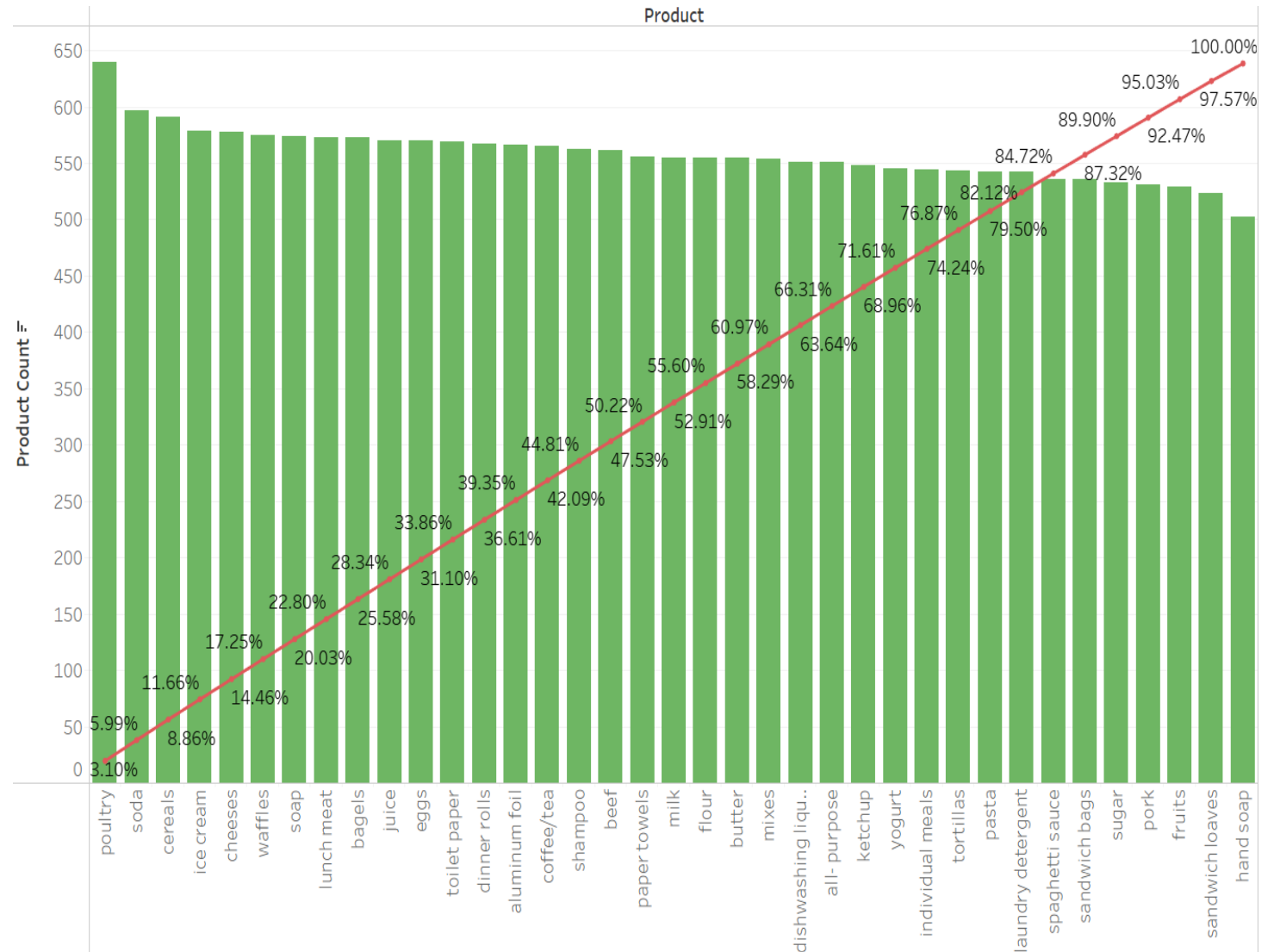


1. UNIVARIATE ANALYSIS

1.2 Pareto Analysis: Top Products Driving Sales

Top 31 products account for 80% of the total sales, which means that top-selling products such as poultry, soda don't dominate the sales as dramatically.

This suggests that the sales are more evenly spread across products, and there's no small set of products overwhelmingly driving the majority of purchases.

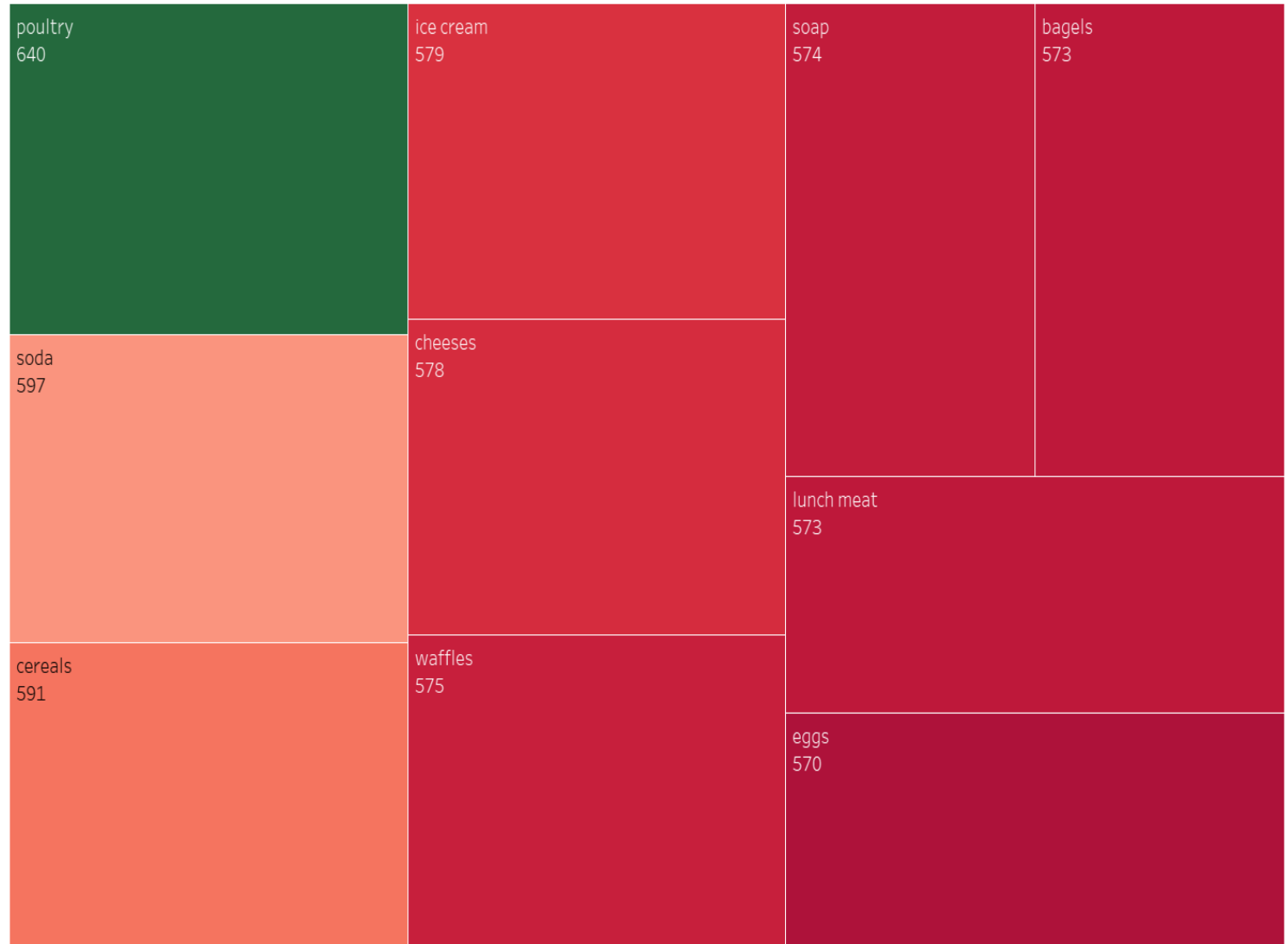


1. UNIVARIATE ANALYSIS

1.3 Top 10 Best-Selling Products

Although 31 products account for 80% of sales, analysing the top 10 products reveals key high-selling categories.

These include poultry, meat, and eggs as one category, ice cream and waffles as another, cereals, bagels, and cheeses as a third, along with soda and soap as separate categories.



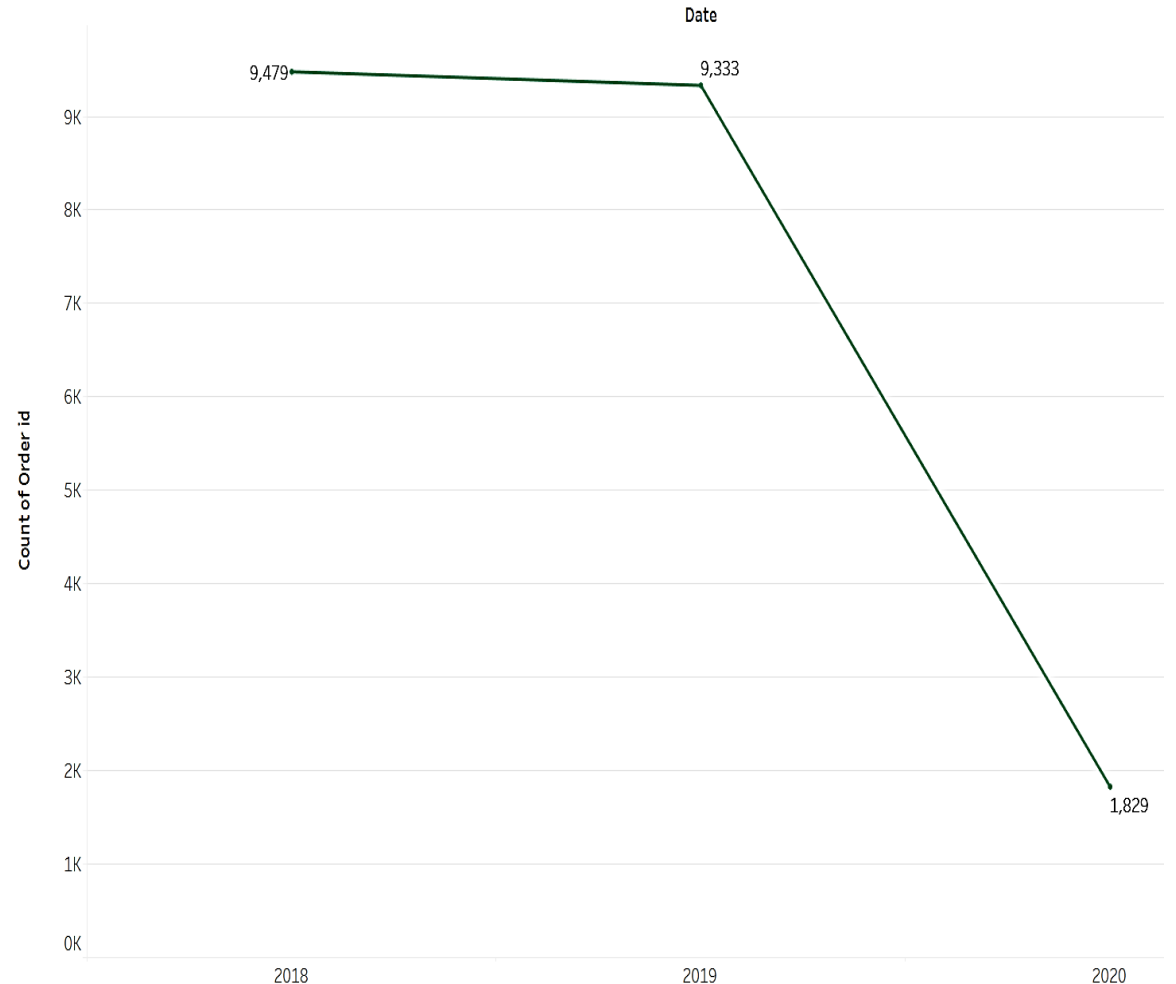
2. BIVARIATE ANALYSIS

2.1 Yearly Transaction Trends

The sales figures for 2018 and 2019 are nearly identical, with 9479 sales in 2018 and 9333 in 2019, slightly favouring 2018.

However, sales in 2020 are noticeably lower, though it's important to note that the data for 2020 covers only two months.

Therefore, further analysis of sales by month across all three years may be necessary.

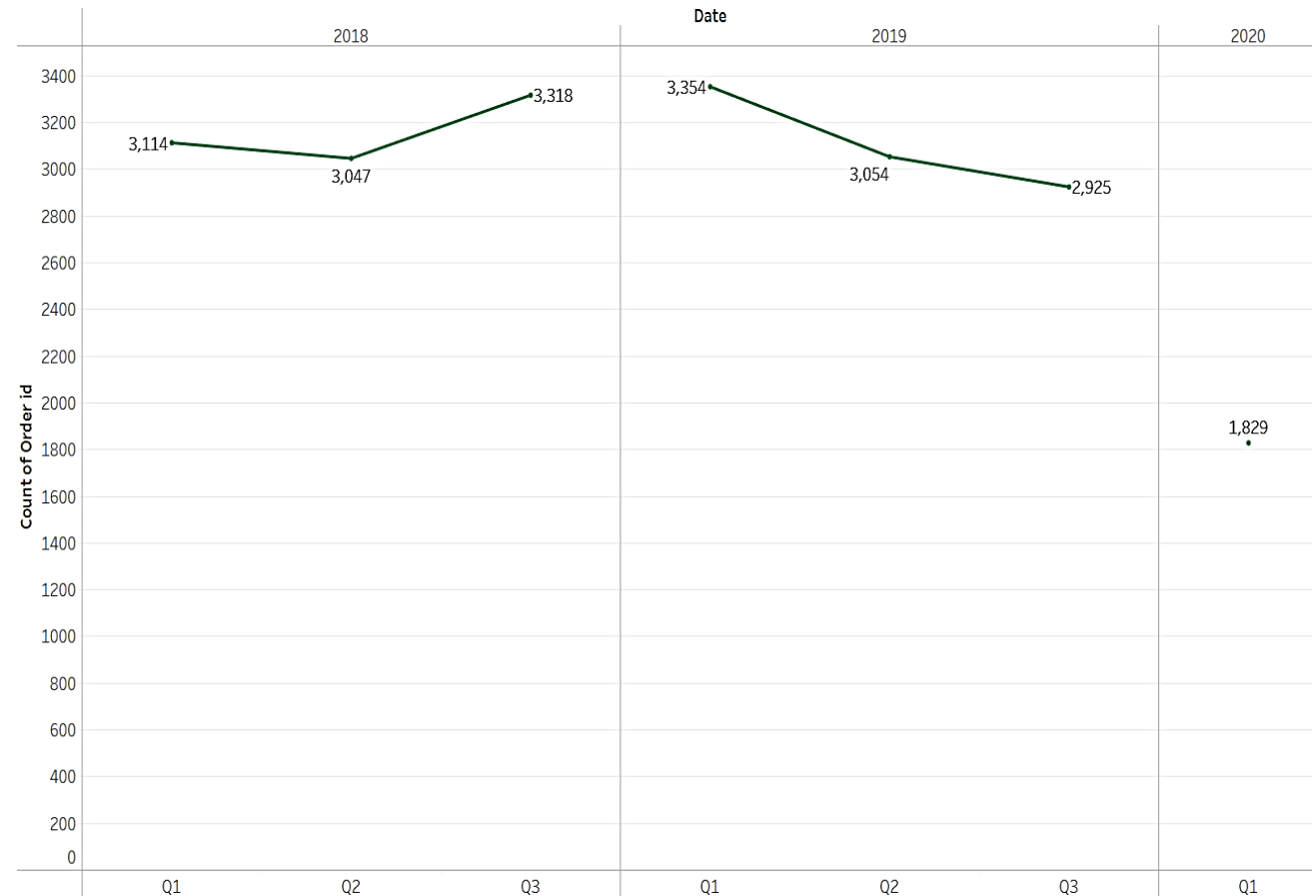


2. BIVARIATE ANALYSIS

2.2 Quarterly Transaction Trends

Q3 shows the highest transaction volume in both 2018 and 2019, while Q2 experiences the lowest sales in both years.

In 2020, sales are the lowest due to data being available for only two months.



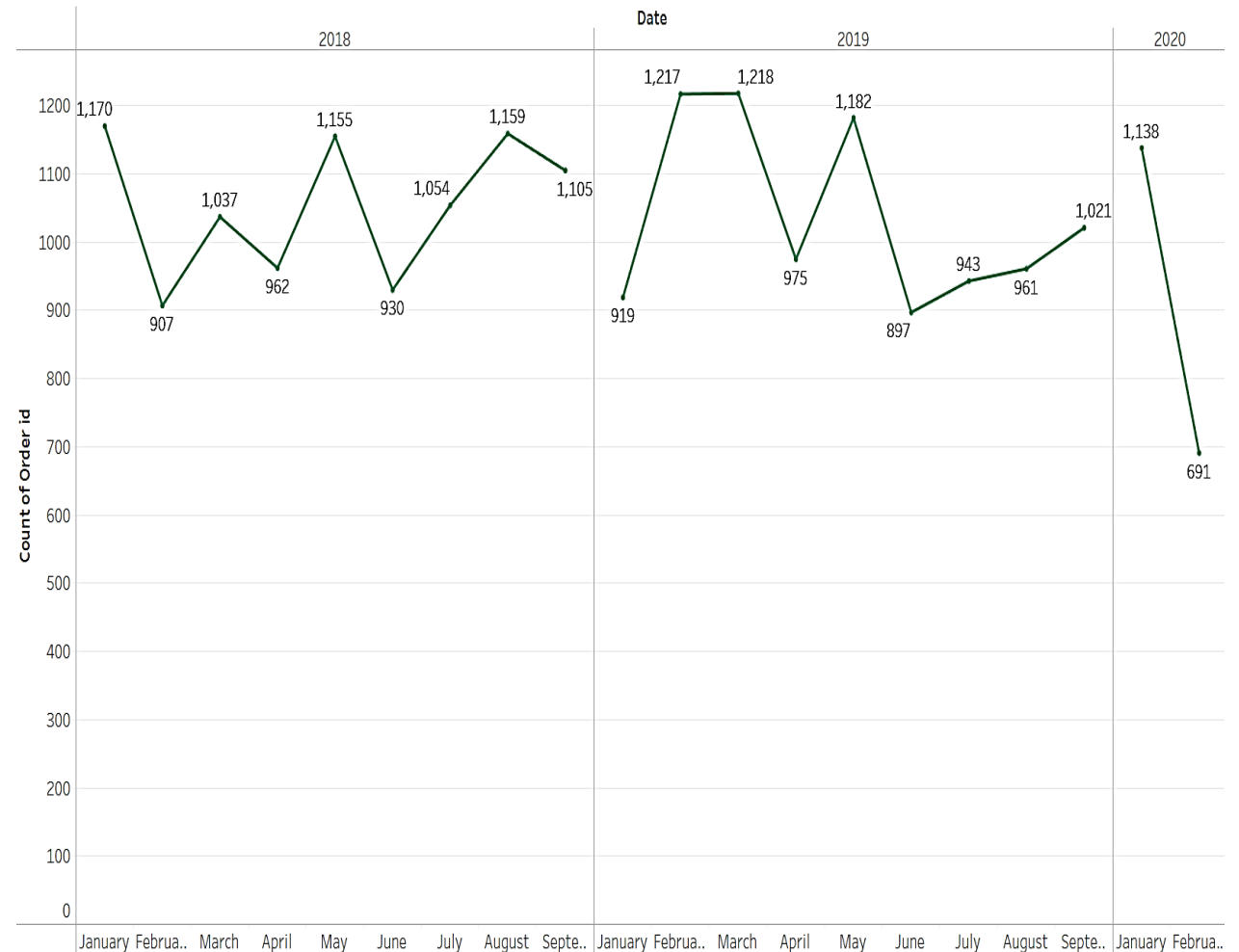
2. BIVARIATE ANALYSIS

2.3 Monthly Transaction Trends

January sees the highest transactions in both 2018 and 2020, while March has the highest transactions in 2019.

A clear seasonality trend is observed in April, May, June, and July, with transaction volume dipping in April, increasing in May, and then decreasing again in June.

However, 2020 experienced a significant decline from January to February, dropping from 1138 to 691 transactions.

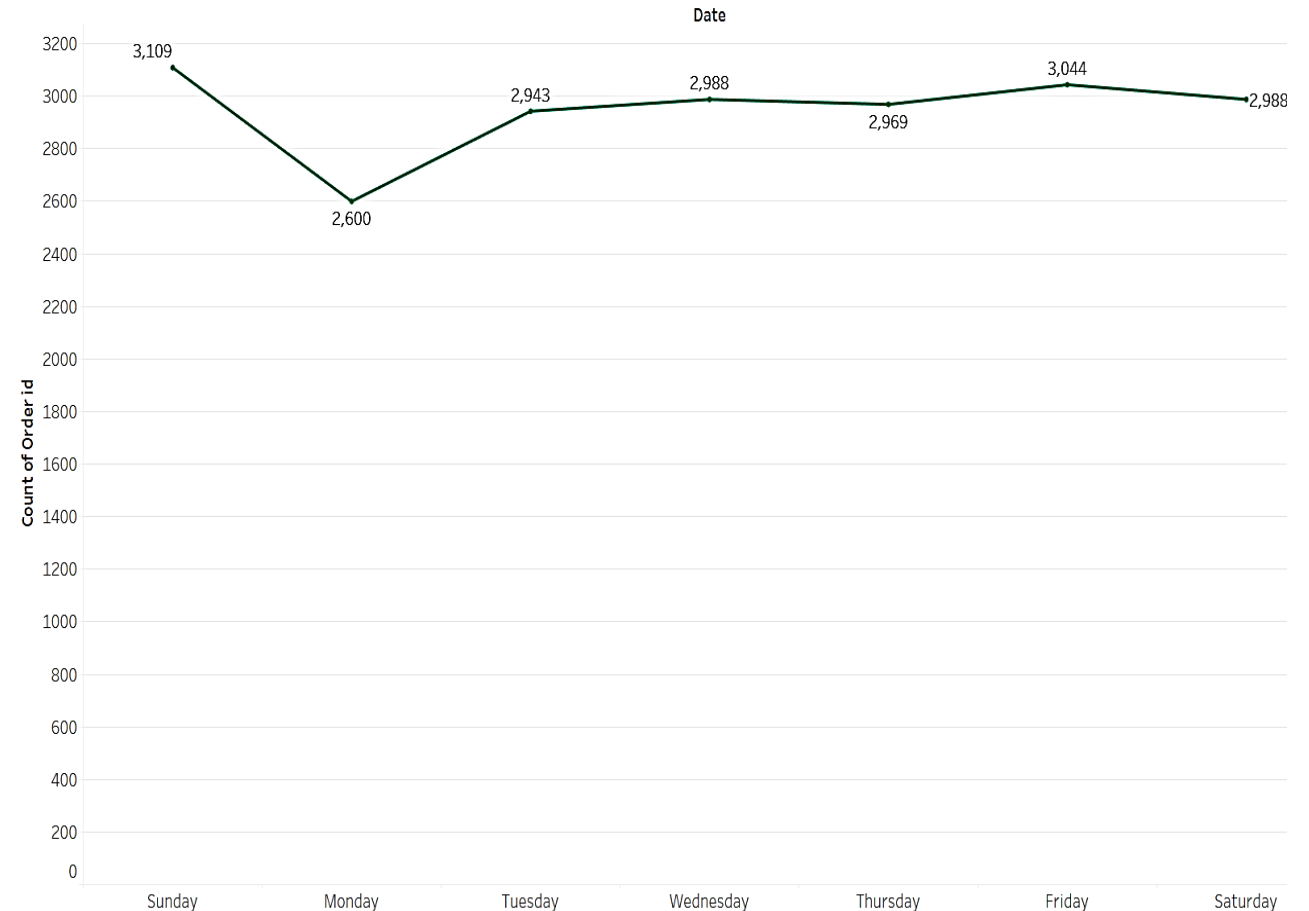


2. BIVARIATE ANALYSIS

2.4 Weekly Transaction Trends

Sunday leads in transaction volume, followed by Friday, while Monday records the lowest number of orders.

This pattern may be attributed to customers making more purchases at the end of the week, possibly due to weekend shopping habits, while Monday sees a drop-in activity as customers return to work and start their week.



2. BIVARIATE ANALYSIS

2.5 Weekly Product Count Across All Years

In 2018, transaction volume is predominantly concentrated on Sundays and Fridays.

However, this pattern diminishes in 2019 and is significantly reduced in 2020, showing a gradual decline over time.

This could be due to changing customer behaviour, shifts in shopping habits, or external factors like the pandemic affecting purchasing patterns in 2020.

Date								Date								Date							
Product	Sund..	Mon..	Tues..	Wed..	Thur..	Friday	Satur..	Product	Sund..	Mon..	Tues..	Wed..	Thur..	Friday	Satur..	Product	Sund..	Mon..	Tues..	Wed..	Thur..	Friday	Satur..
all-purpose	87	82	81	75	73	74	79	all-purpose	36	40	37	37	21	39	37	all-purpose	9	9	7	8	8	5	10
aluminum foil	91	69	84	68	96	86	72	aluminum foil	45	33	41	35	45	37	22	aluminum foil	4	7	2	6	5	7	11
bagels	90	69	95	67	93	88	71	bagels	46	31	43	41	38	42	25	bagels	7	6	5	4	8	7	11
beef	87	67	73	73	81	83	97	beef	40	25	36	34	24	38	38	beef	7	7	5	10	5	6	18
butter	77	79	63	80	89	71	96	butter	35	41	28	45	33	28	41	butter	5	7	7	6	7	6	13
cereals	96	73	84	73	97	93	75	cereals	43	30	34	32	36	42	32	cereals	11	7	4	8	8	5	8
cheeses	84	67	79	81	93	95	79	cheeses	39	37	34	44	33	47	31	cheeses	13	6	3	5	6	6	15
coffee/tea	87	59	98	94	85	75	67	coffee/tea	38	29	47	49	27	33	33	coffee/tea	5	5	3	11	4	8	8
dinner rolls	69	71	91	82	82	88	84	dinner rolls	31	30	50	41	27	38	39	dinner rolls	11	5	6	11	14	12	8
dishwashing liqu..	80	67	70	83	75	89	87	dishwashing liqu..	44	28	37	49	32	42	39	dishwashing liqu..	7	6	3	6	9	7	11
eggs	86	66	83	91	82	86	76	eggs	51	26	37	52	34	35	30	eggs	6	9	5	5	6	8	8
flour	87	75	77	77	73	83	83	flour	34	26	30	42	23	31	38	flour	10	9	4	8	7	8	5
fruits	78	65	64	83	77	82	80	fruits	36	33	29	38	32	31	32	fruits	7	3	1	2	9	8	12
hand soap	67	60	63	78	73	85	76	hand soap	36	21	25	35	28	36	25	hand soap	1	5	5	3	12	9	7
ice cream	87	81	82	81	89	87	72	ice cream	47	32	38	46	36	36	22	ice cream	5	13	2	8	7	13	10
individual meals	86	73	79	81	73	84	68	individual meals	50	26	31	41	23	42	25	individual meals	5	11	2	6	7	6	12
juice	97	74	79	80	75	87	78	juice	51	39	38	43	30	34	31	juice	2	4	5	5	6	8	12
ketchup	87	58	77	91	73	79	83	ketchup	41	25	33	43	21	31	35	ketchup	6	6	7	7	6	7	15
laundry detergent	87	75	66	89	72	70	83	laundry detergent	42	33	34	38	38	31	44	laundry detergent	12	2	2	5	4	7	8
lunch meat	98	69	87	66	85	80	88	lunch meat	46	28	50	30	32	33	37	lunch meat	9	5	2	5	7	8	12
milk	84	67	90	75	80	89	70	milk	44	28	48	46	33	39	32	milk	6	9	4	5	7	6	5
mixes	84	67	73	78	89	88	75	mixes	36	28	27	43	33	42	23	mixes	8	10	4	8	6	6	14
paper towels	80	70	80	68	82	83	93	paper towels	36	28	43	44	36	45	35	paper towels	3	9	4	6	7	7	18
pasta	70	68	74	95	71	72	92	pasta	34	28	33	43	24	32	38	pasta	8	7	2	8	8	4	8
pork	89	68	83	65	74	79	73	pork	49	26	37	33	25	32	27	pork	12	10	5	6	12	9	9
poultry	82	99	82	82	88	102	105	poultry	40	42	33	47	29	46	51	poultry	12	7	9	5	11	9	12
sandwich bags	75	75	76	66	70	84	90	sandwich bags	37	40	30	25	27	39	44	sandwich bags	8	3	9	6	6	6	9
sandwich loaves	81	68	89	76	71	78	60	sandwich loaves	41	31	33	43	27	38	31	sandwich loaves	9	13	5	6	11	9	2
shampoo	88	67	89	91	72	75	80	shampoo	38	31	39	46	29	41	21	shampoo	5	6	4	4	5	5	13
soap	89	76	81	93	72	74	89	soap	45	28	39	47	29	35	42	soap	8	9	7	4	7	5	9
soda	75	75	85	108	80	87	87	soda	34	38	36	65	33	47	34	soda	6	4	6	10	10	4	14
spaghetti sauce	88	62	78	81	77	68	82	spaghetti sauce	36	33	32	49	31	25	39	spaghetti sauce	9	5	5	1	4	8	5
sugar	70	65	81	91	70	81	75	sugar	39	24	41	45	33	40	33	sugar	3	5	5	3	2	4	11
toilet paper	80	66	87	83	83	84	86	toilet paper	38	31	40	41	30	33	45	toilet paper	4	5	8	6	7	7	14
tortillas	81	73	56	80	81	79	93	tortillas	40	36	32	43	27	29	46	tortillas	7	12	2	8	4	8	10
waffles	101	72	82	77	98	76	69	waffles	56	38	34	38	33	33	38	waffles	6	7	6	6	4	5	5
yogurt	84	63	82	86	75	80	75	yogurt	38	34	42	47	27	43	34	yogurt	13	6	6	10	6	5	7

2018

2019

2020

2. BIVARIATE ANALYSIS

2.6 Yearly Product Sales: January and February

Since the data for 2020 is only available for January and February, we will focus our analysis on these two months across all years.

The transaction count is highest in 2018, followed by a decline in 2019, with a further drop in 2020.

Notably, February 2020 shows the lowest sales across most products, which could be attributed to the onset of the pandemic.

Product	2018		Date 2019		2020	
	January	February	January	February	January	February
all- purpose	35	24	23	30	33	23
aluminum foil	35	21	22	35	28	14
bagels	27	32	35	32	32	16
beef	31	32	26	29	30	28
butter	28	21	24	26	34	17
cereals	33	27	28	38	34	17
cheeses	35	29	29	31	27	27
coffee/tea	33	24	30	23	31	13
dinner rolls	29	17	24	38	41	26
dishwashing liquid/detergent	27	32	29	40	34	15
eggs	26	22	28	30	32	15
flour	30	21	19	33	40	11
fruits	25	29	21	30	26	16
hand soap	35	28	25	33	22	20
ice cream	33	24	25	38	37	21
individual meals	30	28	17	26	38	11
juice	36	20	31	28	24	18
ketchup	28	34	28	27	35	19
laundry detergent	34	13	28	34	20	20
lunch meat	32	30	16	38	30	18
milk	26	23	32	42	26	16
mixes	31	26	23	47	36	20
paper towels	26	21	30	38	36	18
pasta	30	22	22	31	31	14
pork	30	28	23	25	39	24
poultry	36	27	23	37	35	30
sandwich bags	24	25	19	34	24	23
sandwich loaves	38	21	30	24	31	24
shampoo	47	28	23	31	27	15
soap	35	16	23	32	27	22
soda	40	18	19	45	38	16
spaghetti sauce	31	21	19	30	23	14
sugar	29	28	25	39	21	12
toilet paper	39	28	21	26	26	25
tortillas	30	22	20	27	35	16
waffles	35	25	33	41	23	16
yogurt	21	20	26	29	32	21

2. BIVARIATE ANALYSIS

2.7 Top 10 products- Weekly Transaction analysis

Bagels show a strong preference on alternate days, indicating that customers purchase them regularly.

Poultry sees the highest transactions from Thursday to Saturday, suggesting it is a popular item towards the weekend.

Soda experiences a mid-week peak, particularly on Wednesdays.

Cheese and cereals are mostly purchased on Thursdays and Fridays, possibly indicating higher consumption as the week winds down.

Lunch meat has the highest sales on Sundays, likely linked to weekend meal preparations.

Product	Date						
	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
bagels	90	69	95	67	93	88	71
cereals	96	73	84	73	97	93	75
cheeses	84	67	79	81	93	95	79
eggs	86	66	83	91	82	86	76
ice cream	87	81	82	81	89	87	72
lunch meat	98	69	87	66	85	80	88
poultry	82	99	82	82	88	102	105
soap	89	76	81	93	72	74	89
soda	75	75	85	108	80	87	87
waffles	101	72	82	77	98	76	69

OBSERVATIONS AND INSIGHTS FROM EDA

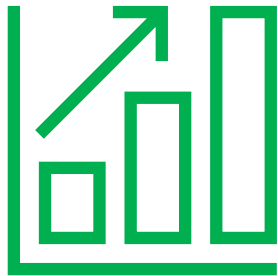


1. Product Distribution & Sales Concentration:

- Poultry leads in transactions with 640, followed by Soda, Cereals, Ice Cream, and Cheeses. However, the sales figures for these products are quite close, indicating a balanced distribution of sales among several popular products.
- While the top-selling products like poultry and soda don't dominate the sales significantly, the fact that 31 products account for 80% of the total sales reveals that a broad array of products contribute to the overall revenue, preventing reliance on a few items.

OBSERVATIONS AND INSIGHTS FROM EDA

2. Sales Trends Over the Years:



- The sales figures in 2018 and 2019 are nearly identical, with a slight edge in 2018 (9479) over 2019 (9333). Sales in 2020 are notably lower, but it's important to note that data for only two months (January and February) is available. The reduced sales in 2020 could be indicative of market disruptions or results of pandemic.
- Both 2018 and 2019 saw the highest transaction volume in Q3, which is significant for planning promotions and inventory for this period and lower sales in seen in Q2.
- The observed seasonal pattern in April, May, June, and July shows a dip in April, an increase in May, and a decline again in June. This could reflect seasonal buying behaviour and help tailor promotions for those months.
- Sunday and Friday consistently lead in transaction volume, likely due to weekend shopping habits. The gradual decline could point to evolving customer shopping habits or external factors like the pandemic or changes in work
- Mondays show the lowest transaction volume, which is likely due to customers returning to work and starting their week, which results in lower shopping activity.

OBSERVATIONS AND INSIGHTS FROM EDA

3. Product-Specific Trends



- Bagels are purchased consistently on alternate days, suggesting they are a regular, frequent purchase for customers. This regularity could indicate potential for targeted promotions to encourage repeat buys.
- Poultry transactions are highest from Thursday to Saturday, likely linked to weekend meal planning. This trend suggests that poultry is a staple item for weekend shopping.
- Soda shows a peak on Wednesdays, suggesting it might be a mid-week treat or a product consumed in larger quantities during the middle of the week.
- Both cereals and cheeses have higher sales on Thursdays and Fridays, possibly indicating an increased demand as the week progresses toward the weekend.
- The highest sales for lunch meat are observed on Sundays, which likely reflects increased meal preparation and family gatherings over the weekend.

MARKET BASKET ANALYSIS

Association Rule Mining (ARM):

It is a powerful technique used in data mining to discover interesting relations between variables in large datasets. The goal of ARM is to find associations or patterns that show how the occurrence of one item is associated with the occurrence of another. It helps to uncover meaningful relationships between products, which can drive more targeted marketing efforts, improve sales strategies, and optimize inventory management.

MARKET BASKET ANALYSIS

Process of Using ARM in Market Basket Analysis:

Data Collection and Preprocessing:

- The first step is to collect transactional data, such as Point-of-Sale (POS) transactions, where each record includes information about the products purchased together by customers in a specific order.
- The dataset is cleaned and transformed to a format that is suitable for analysis.

Defining Parameters (Support, Confidence, Lift)

- Support: Support indicates how frequently the itemset appears in the dataset. Support value of 0.05 (5%) is used in the present analysis which ensures that we are focusing on item sets that appear frequently enough in transactions to be useful for creating combo offers or promotions.
- Confidence: Confidence indicates the likelihood that an item B is purchased when item A is purchased. Confidence value of 0.6 (60%) is recommended for actionable rules which ensures that we get reliable rules that show strong relationships between products.
- Lift: Lift measures the strength of the rule over the expected occurrence of the items independently. A lift greater than 1 indicates that A and B are more likely to be bought together than expected.

MARKET BASKET ANALYSIS

Mining the Association Rules

- ARM is performed using Apriori algorithm. The algorithm discovers all possible combinations of items and generate rules based on the defined thresholds for Support, Confidence, and Lift.
- Rules (0-24) are obtained for the present data set.
- Each rule comprises products that are bought together, product that is likely to be bought with the products that are bought together, support, confidence and lift values.

Evaluating the Quality of Rules

- The rules are filtered based on their Support, Confidence, and Lift values. Higher values of these metrics typically indicate more reliable and actionable rules.
- The present data is set to thresholds, preferably Support = 5%, Confidence = 60%, Lift sorted in descending order (1.791 being the highest) to filter out weak or irrelevant rules and focus on the strongest associations, which is shown below.

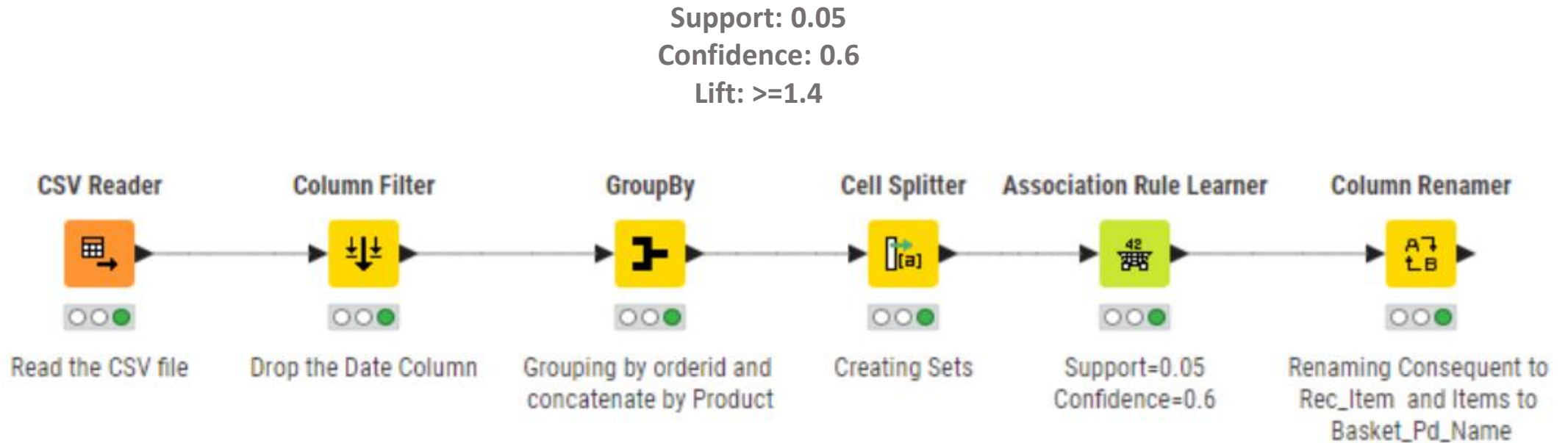
MARKET BASKET ANALYSIS

Generating Insights for Business Actions

- **Product Bundling:** Products frequently bought together can be bundled into promotions or combo packs.
- **Cross-selling:** If customers often buy a specific set of products together, retailers can recommend these products to increase the basket size.
- **Store Layout Optimization:** Products that are frequently bought together can be placed near each other in-store or in online shopping carts to encourage customers to buy more.
- **Targeted Marketing:** Retailers can send personalized offers or discounts to customers based on their buying patterns derived from the association rules.

KNIME WORKFLOW

KNIME is used to perform the RFM Analysis and below is the workflow diagram.



FINAL OUTPUT TABLE

Output Table (Table)

Rows: 24 | Columns: 6

☐	#	RowID	Support Number (double)	Confidence Number (double)	Lift Number (double)	Rec_Item String	implies String	Basket_Pd_Name Set
☐	1	rule0	0.05	0.64	1.7	juice	<---	[yogurt,toilet paper,aluminum foil]
☐	2	rule1	0.05	0.62	1.645	juice	<---	[yogurt,poultry,aluminum foil]
☐	3	rule2	0.05	0.613	1.616	coffee/tea	<---	[yogurt,cheeses,cereals]
☐	4	rule3	0.05	0.6	1.424	poultry	<---	[dishwashing liquid/detergent,laund
☐	5	rule4	0.051	0.63	1.678	mixes	<---	[yogurt,poultry,aluminum foil]
☐	6	rule5	0.051	0.611	1.66	sandwich bags	<---	[cheeses,bagels,cereals]
☐	7	rule6	0.051	0.674	1.726	cheeses	<---	[bagels,cereals,sandwich bags]
☐	8	rule7	0.051	0.617	1.558	cereals	<---	[cheeses,bagels,sandwich bags]
☐	9	rule8	0.051	0.63	1.621	dinner rolls	<---	[spaghetti sauce,poultry,cereals]
☐	10	rule9	0.051	0.637	1.512	poultry	<---	[dinner rolls,spaghetti sauce,cereals
☐	11	rule10	0.051	0.604	1.589	milk	<---	[poultry,laundry detergent,cereals]
☐	12	rule11	0.052	0.628	1.61	eggs	<---	[dinner rolls,poultry,soda]
☐	13	rule12	0.052	0.641	1.649	dinner rolls	<---	[spaghetti sauce,poultry,ice cream]
☐	14	rule13	0.052	0.686	1.628	poultry	<---	[dinner rolls,spaghetti sauce,ice cre
☐	15	rule14	0.052	0.628	1.614	dinner rolls	<---	[spaghetti sauce,poultry,juice]
☐	16	rule15	0.052	0.602	1.429	poultry	<---	[dinner rolls,spaghetti sauce,juice]
☐	17	rule16	0.052	0.634	1.627	eggs	<---	[paper towels,dinner rolls,pasta]
☐	18	rule17	0.052	0.602	1.621	pasta	<---	[paper towels,eggs,dinner rolls]
☐	19	rule18	0.054	0.642	1.651	dinner rolls	<---	[spaghetti sauce,poultry,laundry det
☐	20	rule19	0.054	0.656	1.556	poultry	<---	[dinner rolls,spaghetti sauce,laundry
☐	21	rule20	0.055	0.624	1.565	ice cream	<---	[paper towels,eggs,pasta]
☐	22	rule21	0.055	0.63	1.616	eggs	<---	[paper towels,ice cream,pasta]
☐	23	rule22	0.055	0.643	1.731	pasta	<---	[paper towels,eggs,ice cream]
☐	24	rule23	0.055	0.649	1.791	paper towels	<---	[eggs,ice cream,pasta]

MARKET BASKET ANALYSIS-KEY INSIGHTS

- The market basket analysis aligns well with the exploratory data analysis (EDA) conducted on the dataset.
- The output table highlights item combinations that predominantly feature top-selling products, including poultry, cereals, cheeses, ice cream, bagels, and eggs.
- Focusing on these items in combination offers a strategic opportunity to further boost sales for already high-performing products.
- Additionally, the analysis introduces new items to the list, such as dinner rolls, spaghetti sauce, juice, pasta, and yogurt.
- Incorporating these newly identified products into sales strategies can help increase revenue by creating more appealing product bundles and promotions that resonate with customer buying patterns.

The top combination items to consider include:

- | | | |
|----------------|------|--|
| • paper towels | <--- | [eggs, ice cream, pasta] |
| • poultry | <--- | [dinner rolls, spaghetti sauce, laundry detergent] |
| • poultry | <--- | [dinner rolls, spaghetti sauce, cereals] |
| • poultry | <--- | [dinner rolls, spaghetti sauce, juice] |
| • poultry | <--- | [dinner rolls, spaghetti sauce, ice cream] |
| • cheeses | <--- | [bagels, cereals, sandwich bags] |
| • cereals | <--- | [cheeses, bagels, sandwich bags] |
| • eggs | <--- | [paper towels, dinner rolls, pasta] |
| • juice | <--- | [yogurt, toilet paper, aluminum foil] |

MARKET BASKET ANALYSIS-KEY INSIGHTS

Paper towels-[eggs, ice cream, pasta]

- The combination suggests that when customers purchase eggs, ice cream, and pasta together, they are also likely to buy paper towels.
- It indicates that these items are often bought in the same context, possibly during grocery shopping trips for household and meal-related items.

Poultry-[dinner rolls, spaghetti sauce, laundry detergent]

- The combination suggests that when customers purchase dinner rolls, spaghetti sauce and laundry detergent together, they are also likely to buy paper poultry.
- This could reflect typical meal preparation patterns by preparing poultry with dinner rolls and spaghetti sauce while also buying household cleaning items.

Poultry-[dinner rolls, spaghetti sauce, cereals]

- Similar to the previous combination, this suggests that poultry is frequently bought alongside dinner rolls, spaghetti sauce, and cereals, indicating that these items are often part of a single shopping trip, potentially for a family meal.

Poultry-[dinner rolls, spaghetti sauce, juice]

- The combination implies that customers purchasing poultry are also likely to buy dinner rolls, spaghetti sauce, and juice.
- It could represent a typical meal combination where customers pair poultry with a drink and sides like rolls and sauce.

MARKET BASKET ANALYSIS-KEY INSIGHTS

Poultry-[dinner rolls, spaghetti sauce, ice cream]

- The combination shows that customers buying poultry are also purchasing dinner rolls, spaghetti sauce, and ice cream, suggesting that poultry is part of a larger meal where these sides and desserts are also included.

Cheeses-[bagels, cereals, sandwich bags]

- The combination indicates that customers buying cheeses often buy bagels, cereals, and sandwich bags as well.
- It suggests a preference for breakfast or snack-related products that complement the cheeses, such as bagels or sandwiches.

Cereals-[cheeses, bagels, sandwich bags]

- The combination shows that customers who purchase cereals also tend to buy cheeses, bagels, and sandwich bags.
- This is likely linked to breakfast or lunch combinations, where cereals, cheeses, and bagels are often purchased together.

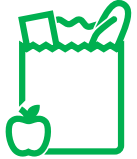
Eggs-[paper towels, dinner rolls, pasta]

- When customers buy eggs, they are likely to also purchase paper towels, dinner rolls, and pasta, which might reflect a meal preparation where eggs are used in cooking alongside other meal items, and paper towels are used for cleaning or food handling.

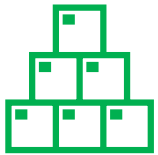
Juice-[yogurt, toilet paper, aluminum foil]

- The combination suggests that juice is frequently bought alongside yogurt, toilet paper, and aluminum foil, possibly indicating a shopping trip where customers are buying both grocery items and household essentials.
- The combination could reflect customers' purchasing behavior when stocking up on diverse products during a single trip.

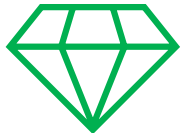
BUSINESS RECOMMENDATIONS



- Based on frequent item combinations, the store can create bundle promotions such as buying eggs, ice cream, and pasta together, and getting a discount on paper towels or buying poultry with dinner rolls, spaghetti sauce, and ice cream, and saving on a meal kit. This can increase average basket size and encourage customers to buy multiple complementary items.

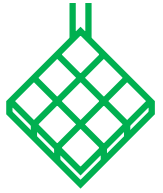


- Given that products like poultry, dinner rolls, and spaghetti sauce are frequently bought together, the store could cross-sell by recommending these items on the same shelf or through online platforms. In-store signage or digital ads can highlight these complementary items.



- As many combinations include items for full meals, such as poultry with dinner rolls, spaghetti sauce, and juice, consider offering meal planning promotions. These could include discounts or loyalty points for customers who purchase full meal kits, helping them streamline their shopping experience while driving sales of multiple items.

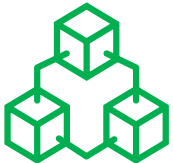
BUSINESS RECOMMENDATIONS



- As seasonality could be seen from EDA, items like ice cream and dinner rolls often appear together, especially in certain seasons (summer for ice cream, holiday seasons for dinner rolls). Offering these items as part of seasonal bundles can help drive sales during peak periods.

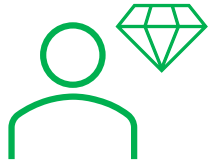


- The combination of juice with yogurt, toilet paper, and aluminum foil suggests a mix of health and household products. This can guide the store to create health-focused or household essentials shopping lists, offering discounts or incentives for customers purchasing both categories.



- Placing related items like cheeses with bagels, cereals, and sandwich bags near each other to make it easier for customers to pick them up in one shopping trip. This can increase the likelihood of multiple-item purchases and boost overall sales.

BUSINESS RECOMMENDATIONS



- Based on recurring item combinations, offer personalized loyalty rewards, like, customers who frequently buy poultry with dinner rolls, spaghetti sauce, and juice could receive a reward when purchasing this combination, reinforcing purchasing behavior.



- The item associations obtained can be used to power recommendation engines on e-commerce platforms. When a customer adds an item like poultry to their cart, suggest related items like dinner rolls, spaghetti sauce, or juice, making the shopping experience more intuitive and increasing the likelihood of additional purchases.

By leveraging the insights from the market basket analysis, the grocery store can enhance the shopping experience, increase sales, and strengthen customer loyalty.